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ChemMate: an autonomous research mate for the full computational chemistry research lifecycle, from ideation to manuscript

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Abstract

Agentic research systems have split along a line. Systems that automate the research lifecycle, from ideation to manuscript, now reach flagship venues, but their experiments are machine-learning runs, biomedical hypotheses, or self-written software, not physical simulation. Systems that drive real computational-chemistry engines stop at the calculation and rarely leave a single cluster. Here we present ChemMate, an autonomous research mate that joins the two for computational chemistry. Riding a general-purpose coding-agent harness, ChemMate adds roughly two hundred expert-protocol skills bound by a machine-checked runtime contract, an idea pipeline spanning applied, method-development, and theory modes, a typed research knowledge graph, and native multi-cluster orchestration designed so that each job leaves a durable, human-readable receipt recording which calculation ran on which machine. Autonomy is bounded by construction: structure-check and submission gates keep a human in the loop exactly where wrong inputs would burn allocation, and the gates are web-actionable, including from a phone, so a researcher can inspect, correct, and approve a job away from the desk. We demonstrate the system through five campaigns spanning electrocatalysis, non-thermal-plasma catalysis, homogeneous asymmetric catalysis, and biomolecular drug discovery, with a concrete result in each: a machine-learned density-functional Hamiltonian whose agent replaced a misleading success metric and cut a band-structure error from 491 to 101 meV; a from-scratch plasma-kinetics model reproducing the measured ethylene-to-acetylene ratio; an enantioselectivity prediction matching the crystallographic assignment at more than 99% enantiomeric excess; a virtual screen returning 100 interference-clean protease hits at 16.5-fold enrichment; and a reconstruction, from a published paper alone, of an isolated-Pd CO-reduction proton-supply mechanism across structural, energetic,

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Abstract

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1 Introduction

Computational chemistry advances through a long chain of human labor that the calculation itself occupies only briefly. A researcher reads the literature, forms a hypothesis, chooses a method, builds structures, prepares and submits jobs across whatever clusters hold an allocation, watches

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them, recovers the ones that fail, checks convergence by force rather than by a scheduler’s exit code, extracts the physics, and finally writes it up. Each link is a place where effort leaks and errors enter, and the calculation an automated agent can now run is a small fraction of the whole. The opportunity is not a faster calculation but an automated traversal of the chain.

Two lines of agentic research have approached this opportunity from opposite ends, and they have not met. The first automates the *reasoning* loop of research. Sakana’s AI Scientist chained idea generation, experimentation, writing, and automated review into one closed pipeline [1]; Google’s Co-Scientist organized hypothesis generation as a multi-agent tournament under a scientist-in-the-loop, with biomedical hypotheses confirmed in cell-line assays [2]; FutureHouse’s Robin nominated a dry-AMD therapeutic validated at the bench [3]; and Google’s empirical-software system discovers expert-level analysis methods by tree search against a scorable metric [4]. Earlier and lighter-weight systems established the template: Data-to-Paper foregrounded back-traceable information flow from dataset to manuscript claim [5], AutoSci added a typed memory graph and a self-evolution loop spanning literature to rebuttal [6], and scienceclaw coordinated decentralized agents through an immutable artifact-lineage DAG [7]; a position paper frames the agenda these systems pursue [8]. Ideation-to-manuscript automation now appears in flagship journals, so the lifecycle axis is no longer open territory. The scientific substrate of these systems, however, is machine-learning research, biomedical hypothesis generation, or software. None runs a density-functional-theory (DFT), molecular-dynamics (MD), or non-adiabatic-dynamics campaign, and their experiments execute on local hardware or abstracted compute rather than on scheduler-managed clusters.

The second line automates the *execution* loop of computational chemistry, driving real simulation engines on real hardware. QUASAR pairs a Strategist–Operator–Evaluator agent triple with a benchmark spanning routine to frontier atomistic tasks, but treats job submission as an explicit non-goal [9]; OpenClaw layers reusable domain skills and a dispatcher onto a general coding agent and does submit to heterogeneous schedulers through one unified interface, demonstrated on a single methane-oxidation workflow [10]; AtomisticSkills, the closest architectural relative, curates a library of more than a hundred skills on a general coding agent, with remote DFT bound to one institutional cluster [11]; AICC, the closest in governance machinery, couples its skill library to a machine-readable control plane, task nodes, append-only logs, execution leases, four review gates, and evaluates its agent by replaying real referees’ computational requests against the authors’ published answers, on one cluster, with approval gates that relax to documented defaults when unattended [12]; and DREAMS adds a dedicated convergence agent and a shared canvas memory, reaching expert-level adsorption-energy accuracy on a single SLURM cluster [13]. The El Agente family explores agent topology, moving from a hierarchical-memory multi-agent quantum-chemistry system [14] to typed execution graphs on a single agent that outperform the multi-agent predecessor on the same benchmark [15], and has since specialized into quantum-simulation, solid-state, molecular-editing, research-collaborator, and agent-forging variants [16–20]; SciToolAgent retrieves tools from a manually built knowledge graph with a safety module [21], and a prompt-engineered simulation agent sustained twenty self-validated cycles on a polymer-conformation task [22]. These named exemplars sit inside a rapidly expanding ecosystem: agents that drive individual engines and simulation workflows [23–29]; agents aimed at specific discovery tasks, from adsorption configurations, metal–organic frameworks, and materials design to computational catalysis, molecular dynamics, and characterization with theory in the loop [30–37]; and agents that compose chemistry tools for reasoning and hypothesis generation [38–43]. Recent reviews map this landscape [44, 45]. These systems establish that language-model agents can drive real chemistry engines, and several reach a scheduler. Yet each stops at execution and analysis: none carries a project onward to a manuscript, and none maintains a project-level, multi-cluster record of which calculation ran on which machine.

A third line closes the discovery loop in the physical world rather than in simulation: a GPT-4

planner driving a cloud laboratory [46], the autonomous A-Lab synthesis facility [47], the Virtual Lab’s multi-agent design of experimentally validated nanobodies [48], biofoundry-coupled enzyme engineering [49], state-machine-grounded gene-editing design [50], human–AI–robot collaboration in catalysis [51], and combined physics-simulation and machine-learning agents for protein design [52]. Benchmarking in that setting warns that strong question-answering performance does not transfer to instrument control and that agents can “sleepwalk” past safety constraints [53]; we take both warnings as design constraints. So too the critical literature: the field’s reviews identify the absence of standard, task-grounded benchmarks as a central gap [54, 55]; a skeptic’s position paper argues that current agents are strong co-scientists but not autonomous discoverers, and calls for simulators as verifiers and persistent world models [56]; and textual-gradient optimization shows what self-improvement looks like when it is left ungoverned [57]. We answer the benchmark critique with an evaluation run on real tasks and real clusters, and the governance concern with self-improvement kept under explicit human review.

The three lines leave an intersection empty. The first automates the reasoning loop but not the simulation; the second automates the simulation but not the lifecycle; the third closes its loop in the laboratory, not on the clusters where computational campaigns run. No system joins full-lifecycle coverage to real computational-chemistry physical simulation *and* to multi-cluster orchestration with a persistent record of which job ran where. The gap matters because real computational-chemistry research lives precisely at that intersection: the science is physical simulation, the simulation runs on high-performance computing that is heterogeneous and spread across institutions, and the value of a campaign is realized only when it ends in a written, defensible result rather than a directory of output files. A system that automates the reasoning loop without the simulation cannot do the chemistry; a system that automates the simulation without the lifecycle leaves the researcher with the same downstream burden; and either, without an audit trail spanning clusters, cannot make a multi-cluster campaign replayable.

In this work we present ChemMate, an autonomous research mate that occupies this intersection. It rides a general-purpose coding-agent harness and supplies what the harness lacks for chemistry. We frame ChemMate not as the first full-lifecycle agent, an axis now occupied, but as the computational-chemistry and high-performance-computing instance of the now-validated class of full-lifecycle research agents, the corner of that class grounded in real physical simulation.

ChemMate is organized as four layers on the harness, tied together by a typed knowledge graph (Fig. 1). The harness supplies file editing, shell execution, and tool invocation but knows nothing of chemistry; ChemMate supplies the chemistry and the discipline. Its research-wiki substrate, the typed-page schema, the graph store, and the skeleton of the idea pipeline, builds on the open-source AutoSci platform [6]; the chemistry is native, from the protocol skills and engine layer to the multi-cluster orchestration, receipts, and gates, which build on our openly released CatGo workbench [58, 59]. The full inherited-versus-native inventory ships alongside the code (Methods).

The first layer is a library of roughly two hundred markdown skills, each encoding one protocol a computational chemist would otherwise carry in their head, assembling a multi-species pseudopotential file, preparing a nudged-elastic-band calculation, extracting a free energy with zero-point and entropic corrections, or running a non-adiabatic-dynamics step, loaded on demand rather than held in context at once. Engines are not hard-wired: VASP [60], Quantum ESPRESSO [61], CP2K [62], ORCA [63], Gaussian [64], ABACUS [65], BDF [66], xTB [67], GPAW [68], SIESTA [69], ABINIT [70], and DFTB+ [71] are each reachable through their own skills, alongside machine-learning potentials such as MACE [72], DeePMD [73], and FairChem [74]. The engine remains the researcher’s choice: the system recommends one from the stated intent and the target cluster’s availability, then waits for confirmation. A machine-checked runtime contract keeps the library coherent: a validator enforces declarative schemas over every page, edge, and field write. Skills evolve only through a

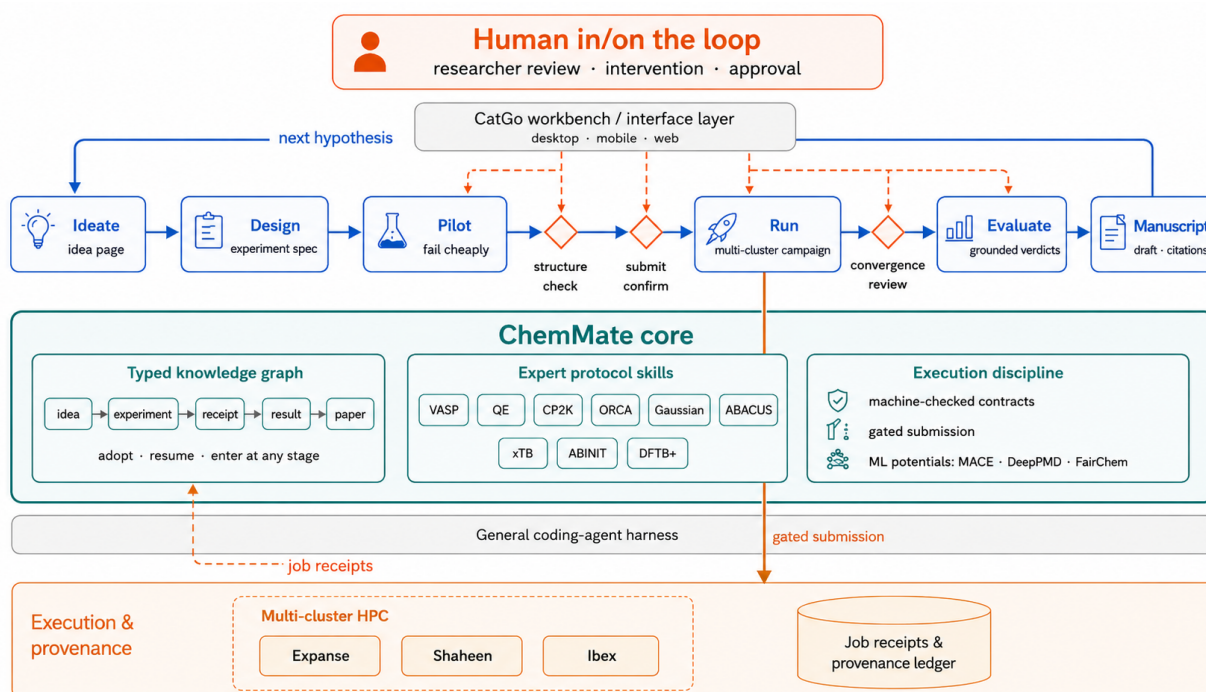


Figure 1: Architecture of ChemMate. A computational-chemistry project travels the full research lifecycle (blue spine, C1): ideation, experiment design, pilot, multi-cluster campaign, grounded evaluation, and manuscript, with evaluation feeding the next hypothesis. The researcher reviews and steers every stage (person marks); the red diamonds are the hard, blocking gates. Expert-protocol skills under a machine-checked runtime contract and a typed knowledge graph (bands) support every stage on top of a general coding-agent harness: ingested literature enters the graph as typed pages, ideation draws on the accumulated graph (optionally seeded with a direction and papers), and every stage reads and writes it. Because all state lives in typed pages and receipts on disk, the lifecycle has no mandatory entry point: an existing project is adopted, and a paused one resumed, at any stage. Campaigns execute on real multi-cluster high-performance computing (orange, C2), where every job leaves a receipt aggregated into the `_REMOTES` ledger and wired back into the graph, making every reported claim reconstructable. Human gates (red diamonds, C3), structure check, submission confirmation, and an adversarial convergence check, are hard-wired into the execution path and remain actionable from any device (phone, tablet, or desktop) through the cross-platform CatGo client.

human-reviewed proposal step, never by automatic self-edit (Methods).

The second layer is an idea pipeline. Research enters the system as an idea, not as a calculation request, and runs through five stages (ideation, experiment design, pilot run, full run, and evaluation), each writing a typed page that the next stage consumes. Ideation is gap-driven rather than prompt-driven. The researcher may seed a round with a direction and papers, but without a seed, ideas are generated against the accumulated knowledge graph, its open questions, the recorded limitations and assumptions of ingested methods, and a banlist of previously failed ideas, along five structured generation paths, by two language models brainstorming independently before their candidates are merged. Every surviving idea must name the reference it will be judged against: an experimental value, a benchmark plus a baseline, or the limiting case a derivation must recover. The researcher selects from the ranked survivors (Methods). Each idea also carries a research mode that branches the design: applied property prediction and screening, method development against a reference benchmark, or pure theory and derivation. The same pipeline therefore serves an overpotential screen, a functional benchmark, and an analytical argument without special-casing, and pilot runs exist to fail cheaply before a campaign commits allocation.

The third layer is native multi-cluster orchestration, the point where ChemMate departs most sharply from systems that stop at the calculation. Its organizing principle is that a project must remain legible and replayable no matter how many jobs it spawns or how many clusters it spans: calculation directories are named for their science rather than hashed; every job leaves a durable receipt recording which machine and remote directory ran it; receipts aggregate into a project-level ledger that survives changes of cluster or configuration; and a multi-candidate screen submits through a batch path that keeps the index readable at hundreds of jobs. The layer monitors running jobs, advances each finished candidate independently so a slow member never blocks the rest, and routes classified failures into bounded, receipt-tracked recovery (Methods). The lifecycle also has no mandatory entry point. Because every stage reads its state from typed pages and receipts on disk rather than from a session’s memory, ChemMate can take over a research effort at whatever stage it has reached. An existing project that predates the system, scattered directories, opaque names, calculations spread across clusters, is adopted into the layout, its calculations indexed under readable names and its remote directories reorganized to mirror the local tree through a plan-then-approve tidy step. A project paused at any stage is resumed from disk by a fresh session, because the narrative files that describe its state travel with the tree (Methods).

The fourth layer makes the human-in-the-loop stance *operational* rather than nominal. The costliest mistake in high-performance computing is a wrong structure submitted at scale, so every structure-related operation and every submission is gated: a pre-execution hook hard-blocks the mutating call until the researcher confirms; cancellation is keyed to explicit job identifiers only; and a finished calculation is not trusted on the scheduler’s word, an adversarial verifier judges convergence by force, given only the bare claimed verdict so that it cannot be anchored. The gates are actionable from anywhere: a cross-platform client (phone, tablet, or desktop) unifies structure visualization, live structure and input-file editing over a remote connection, and gate approval on one surface, so supervision costs no more than the time it takes to glance at a phone (Methods).

The typed knowledge graph binds the layers together. Ideas, experiments, results, methods, concepts, people, and papers are typed pages; edges carry direction, attributes, and confidence; and citations are first-class records. An idea page points to the experiment it spawned, which points to the receipts of the jobs it ran, which point to the results that feed the manuscript draft, so the graph and the receipts together form an audit trail from which any reported result can be reconstructed.

Three contributions define the work. First (C1), ChemMate is, to our knowledge, the first autonomous simulation-agent system to span the full research lifecycle for computational chemistry, carrying a project from an idea page through an experiment-design specification, a high-performance-

Table 1: Campaign overview. Five campaigns across four domains and two research modes (A = applied screening, B = method development). The two molecular/biomolecular cases (nickel, Mpro) are closed.

Campaign	Mode	Scale	Outcome vs. reference	Autonomy
ML density-functional Hamiltonian (HER)	B	DeepH-E3 on a semiconducting electrocatalyst	agent replaced a misleading matrix-element metric with a band-error metric; Fermi-level error 491/715 $\rightarrow$ 101/179 meV	self-audited and corrected its own success metric (C1, C3)
Non-thermal-plasma ethane	A/B	41-species 0D engine + 8 metals	C ₂ H ₄ /C ₂ H ₂ 1.47 vs. 1.35–1.5 exp.; manuscript	built a missing gas-phase skill
Ni asymmetric alkylolation	A	4 channels, 80-atom complex, ORCA	S favored, $\Delta\Delta G^\ddagger = 7.4$ kcal mol ⁻¹ , matching the crystallographic (S) assignment and > 99% ee; the source literature reports no computed TS; IRI quantifies its qualitative steric model	TS located from a given intermediate; found the more-stable N-coord. isomer and showed by barrier that it is not the reactive one
Mpro virtual screen	A	50k retrospective + 50k prospective, docking	AUC 0.795, EF@1% 16.5 $\times$; 100 interference-clean novel hits	full lifecycle; honest self-triage
Isolated-Pd CO reduction	A	Cu(111) $\pm$ Pd, explicit-water constant-potential AIMD	proton-supply mechanism reconstructed across three scales; water-dissociation E_a 1.41 vs. 1.73 eV at the Pd site	mechanism rebuilt from the source paper alone, structure/energetics/electronics in one figure

computing campaign, and a typed knowledge graph to a manuscript draft, with concrete artifacts at every stage. Second (C2), it performs real multi-cluster high-performance-computing orchestration, batch submission, monitoring, automatic advancement of converged candidates, and fault tolerance, backed by a persistent receipts ledger that records which calculation ran on which machine across institutions, a ledger no surveyed system produces; the closest relative’s append-only logs remain per-project and single-cluster [12]. Third (C3), it makes the human-in-the-loop and human-on-the-loop stance operational rather than nominal: structure-check and submission gates, a cancellation safety rule keyed to explicit job identifiers, and an adversarial convergence verifier are architecture, not advice: the gates hard-block even when no one is watching, where the closest relative’s approval gates relax to documented defaults [12], and they are actionable from any device. The Results section supports these claims with five end-to-end campaigns run on real clusters, spanning electrocatalysis, non-thermal-plasma catalysis, homogeneous asymmetric catalysis, and biomolecular drug discovery; the Discussion states the limits of the design; implementation details are given in Methods.

2 Results

We demonstrate ChemMate through five end-to-end campaigns drawn from the system’s own research record (Table 1). The campaigns are chosen for spread rather than for a single theme: they cross four problem domains, electrocatalysis, non-thermal-plasma catalysis, homogeneous asymmetric catalysis, and biomolecular drug discovery, and two research modes, applied screening and method development. This spread is the generality argument. A calculation-only gallery can show one harness driving many engines; what no prior system has shown is the same agent carrying a project from an idea, through a multi-cluster campaign, to a written result, *including* the development of a new method

and an adversarial self-check on its own conclusions. Each campaign lights up a specific contribution as its headline: a machine-learned density-functional Hamiltonian whose agent audited and corrected its own misleading success metric (C1, C3) on the hydrogen-evolution electronic-structure study; harness extensibility on the plasma model; saddle-point location and selectivity attribution from a supplied intermediate, with a self-initiated coordination-isomer check (C1, C3), on the nickel enantioselectivity case; a validate-then-discover virtual screen on the SARS-CoV-2 main-protease case; and a from-the-paper-alone reconstruction of an isolated-Pd CO-reduction proton-supply mechanism across structural, energetic, and electronic scales. The campaigns are at different stages of completion, several have closed with written results or a manuscript draft, while others are still accumulating their final screening numbers, and we report the achieved result and the lifecycle traversal honestly in each case, marking numbers that are still being finalized.

2.1 Evaluation on real campaigns

A recurring criticism of agentic chemistry systems is that they are measured on question-answering proxies rather than on the tasks they claim to perform [9]. We address that criticism directly: our evaluation is the set of end-to-end campaigns themselves, run as real multi-step work, submission, monitoring, failure recovery, convergence checking, and analysis, on real scheduler-managed clusters rather than on a benchmark of isolated questions. Each campaign is scored not by a single number but by whether it clears the bar of a usable scientific result, validated against experiment or a strong literature golden check rather than a reproduction, with every reported number replayable from the shipped receipts and knowledge graph. The five campaigns below are that evaluation; the machine-learned density-functional Hamiltonian for hydrogen-evolution active sites is the worked example of a campaign that cleared this bar and caught its own analysis error, in its own converged output, before reporting.

2.2 A machine-learned density-functional Hamiltonian for hydrogen-evolution active sites, with an autonomous metric-validity audit

Motivation. The activity of a hydrogen-evolution catalyst is read from its electronic structure, the band structure, the projected density of states at the active site, the position of the metal d-band, yet resolving that structure at every candidate site of a large defective surface is exactly where density-functional theory becomes unaffordable. A machine-learned potential returns energies and forces but no wavefunction, so it cannot supply these electronic descriptors. This campaign therefore learns a different object, the density-functional Hamiltonian itself, on a semiconducting electrocatalyst (2H-MoS₂: a ~ 1.2 eV-gap basal plane with metallic edge and sulfur-vacancy active sites) where an equivariant machine-learned Hamiltonian had not previously been landed. It is a method-development campaign (research mode B), and it is this work’s only case whose product is electronic structure rather than an energy, a force, or a continuum term.

Agentic workflow and lifecycle. From an idea and experiment-design page, the system generates density-functional single-point labels (OpenMX), trains an equivariant machine-learned Hamiltonian (DeepH-E3 [75]) against them, and derives site descriptors and the hydrogen adsorption free energy from the predicted spectrum, with a fine-tuned foundation potential supplying the geometries the Hamiltonian model cannot. The scientifically load-bearing episode is an autonomous audit of the system’s own success metric. The first models minimized the Hamiltonian matrix-element error and reached 17.94 meV, beating a four-member local-frame committee (24.77 meV) by 28%, and yet the predicted bands near the Fermi level were wrong by 491–715 meV. The agent diagnosed

that the headline metric was misleading (a low matrix-element error does not guarantee a correct spectrum) and self-corrected by adding a band-aware eigenvalue penalty to the loss; an over-strong version of that penalty then inflated the density of states at the Fermi level and was tuned back down, a correction logged like the rest. Two honest negatives were recorded and kept along the way, an equivariance speed-up that came in at $\sim 1.8\text{--}2.0\times$ rather than the larger factor hoped for, and a Mo-d block-weighting scheme that did not help.

Results. The band-aware loss repaired the electronic structure (Fig. 2): the Fermi-level band error fell from 491/715 to 101/179 meV (in a ± 2 eV window), the d-band-center error from 150/125 to 14.4/7.5 meV, and a coin-flip artifact in the density of states at the Fermi level was eliminated. On four held-out sites the recovered site descriptors correlate with the hydrogen adsorption free energy at Pearson $|r| = 0.89\text{--}0.97$. The payoff is scale: a 386-site screen on a 578-atom defective nanoribbon ran with zero new density-functional calculations, and a query-by-committee active-learning round rescued the small-cell-to-large-cell transfer of the sulfur- p descriptor from Pearson -0.29 to $+0.74$, lifting recall of hydrogen-evolution-active sites from 0% to 78% in the top forty. The campaign is also epistemically explicit about where the method stops: it claims only *relative* adsorption free energies (a $+1.1$ eV absolute offset is disclosed), the d-band-center descriptor transfers across cell size ($+0.743$) while the density of states at the Fermi level does not (-0.024), and the $^*\text{H}$ fine detail is flagged as data-starved. A compiled manuscript reports the method and, as a first-class result, its reliability boundary.

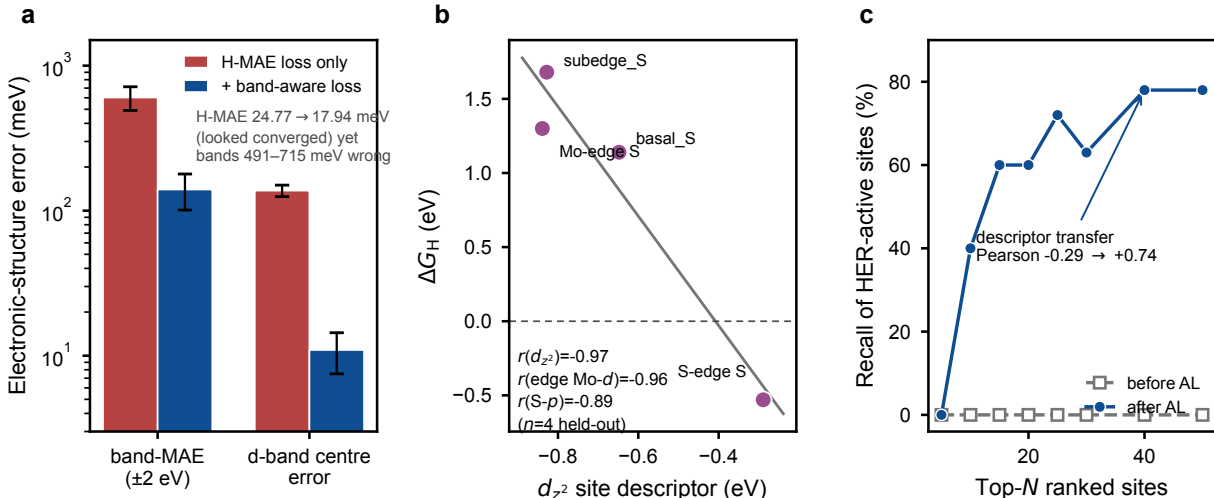


Figure 2: A machine-learned density-functional Hamiltonian for MoS₂ hydrogen evolution, audited against its own success metric. **a**, The Hamiltonian matrix-element error (H-MAE) improved from 24.77 to 17.94 meV yet left the near-Fermi bands wrong by hundreds of meV; a band-aware eigenvalue penalty then cut the band error and the d-band-centre error by roughly $5\times$ and $12\times$ (means over two held-out snapshots; bars, log scale; whiskers span the two snapshots). **b**, The recovered site descriptors correlate with the hydrogen adsorption free energy on held-out sites ($|r| = 0.89\text{--}0.97$, $n = 4$). **c**, A 386-site screen on a 578-atom defective nanoribbon with zero new density-functional calculations: query-by-committee active learning rescues the small-cell-to-large-cell transfer of the sulfur- p descriptor (Pearson $-0.29 \rightarrow +0.74$), lifting recall of hydrogen-evolution-active sites from 0% to 78% in the top forty.

2.3 Multiscale modelling of non-thermal-plasma ethane conversion

Motivation. Non-thermal plasma can drive selective chemistry that thermal catalysis cannot, but predicting it means coupling a non-equilibrium electron-energy distribution to surface kinetics and to first-principles energetics, three scales that are usually handled by three separate communities. This campaign targets the plasma-catalytic semihydrogenation of acetylene from a dilute ethane feed, asking what controls the selectivity to ethylene, and is a test of harness extensibility: the gas-phase scale had no existing skill, so the agent built one.

Agentic workflow and lifecycle. The genuinely new method here is a from-scratch zero-dimensional gas-phase plasma-kinetics engine for ethane chemistry, forty-one species, a reaction set retargeted from methane chemistry, a homebrew Boltzmann electron-energy-distribution solver, and a power-driven reduced field solved self-consistently from the deposited power density rather than set by hand. This engine is coupled, one way, to a surface microkinetic model over six fcc(111) metals and to density-functional descriptors computed across eight metals (seven fcc(111) surfaces plus hexagonal-close-packed Ru(0001)): RPBE+D3(BJ) adsorption energies, d-band centers, crystal-orbital Hamilton-population bond strengths, and nudged-elastic-band barriers with a Brønsted–Evans–Polanyi relation. Building the gas-phase engine and wiring it to the existing surface and density-functional skills is itself the demonstration that the harness extends to a capability it did not ship with, echoing the wrapper-building pattern elsewhere in this work.

Results. The model reproduces an experimental benchmark (Fig. 3; ethylene-to-acetylene ratio 1.47 at the operating point vs. 1.35–1.5 measured) and predicts a clear design target: ethylene yield peaks at a specific energy input of about 1.0 kJ L^{-1} (yield 6.25%, conversion 27.5%). The metal trends correlate cleanly with d-band descriptors, d-band center versus acetylene adsorption energy $r = -0.88$ across the eight metals, a Brønsted–Evans–Polanyi relation between barrier and reaction energy ($R^2 = 0.96$), and crystal-orbital Hamilton-population bond strengths tracking the trend qualitatively, consistent with an electronic origin, and the campaign’s diagnostic is that in this regime power, rather than the catalyst, appears to set selectivity. A venue-ready manuscript exists. We frame the novelty honestly: the plasma chemistry rests on a standard modelling framework, and the method contribution is the from-scratch ethane engine and the multiscale coupling, with the over-prediction of C_3/C_4 byproducts disclosed as a known residual.

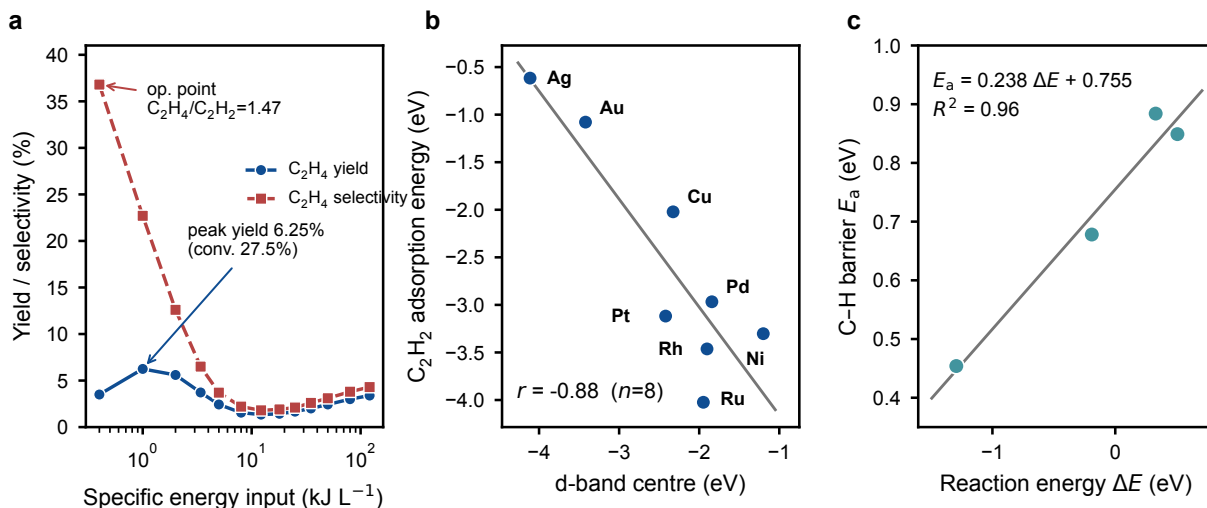


Figure 3: Non-thermal-plasma ethane semihydrogenation: power, not the catalyst, sets selectivity. **a**, Ethylene yield and selectivity against specific energy input from the zero-dimensional plasma-kinetics engine; yield peaks near 1.0 kJ L^{-1} (6.25%, at 27.5% conversion) while selectivity falls monotonically across the sweep, and the operating point reproduces the measured ethylene-to-acetylene ratio of 1.47. **b**, Density-functional descriptor across eight metals: the acetylene adsorption energy tracks the d-band centre ($r = -0.88$, $n = 8$; the hexagonal-close-packed binders Rh and Ru included). **c**, Brønsted–Evans–Polanyi relation between the C–H barrier and reaction energy ($R^2 = 0.96$). Energetics at the RPBE+D3(BJ) level; all values are the campaign’s own outputs.

2.4 Enantioselectivity of a nickel-catalyzed asymmetric radical alkylation

Motivation. Enantioselectivity in transition-metal radical catalysis is decided by a small free-energy difference between diastereomeric transition states, so predicting it exercises the whole chain of the pipeline at once: conformer sampling, saddle-point location, composite thermochemistry, and a selectivity-origin analysis. The system is a nickel-catalyzed enantioselective decarboxylative difluoromethylation [76] that installs a difluoromethylated stereocenter (an 80-atom complex, $\text{C}_{35}\text{H}_{34}\text{F}_2\text{N}_3\text{NiO}_5$, doublet; Fig. 4b). Its mechanism is established: a radical capture / nickel–carbon coupling (step 1) precedes a carbon–carbon reductive elimination (step 2). For nickel-catalyzed radical cross-couplings of this class the capture is fast and reversible while the reductive elimination is enantiodetermining [77, 78], so the campaign models step 2 and does not attempt to locate the step-1 saddle. This case moves the system furthest from surface catalysis, into molecular, homogeneous, stereoselective chemistry, on an engine stack (ORCA, CREST, Multiwfn) shared with no other campaign here. The researcher supplies the oxygen-coordinated reaction intermediate, in which the benzamide binds nickel through its carbonyl oxygen; the workflow carries it through transition-state location, free energies, a coordination-isomer check, and the origin of the selectivity.

Agentic workflow and lifecycle. From an idea and experiment-design page the system scaffolds the project under its own verified cluster preset and traverses the lifecycle unattended (Fig. 4a). From the supplied oxygen-coordinated intermediate it runs a GFN2-xTB conformer search, the diverse subset chosen by energy-windowed farthest-point sampling on heavy-atom root-mean-square deviation, so that rotamers do not masquerade as conformers, and optimizes the reductive-elimination saddle on each of the four $R/S \times cis/trans$ channels. Saddle searches were initialized from the conformer ensemble, except in one channel, where a previously computed in-house geometry was re-optimized at the same level. Minima and saddles are optimized, with harmonic frequencies, at

B3LYP-D3(BJ)/def2-SVP with a CPCM(THF) continuum (RIJCOSX, def2/J auxiliary basis) in ORCA 5.0.1 on the charge-neutral doublet; single points use def2-TZVP/CPCM(THF), and the reported free energies are composite, $G = E[\text{def2-TZVP/CPCM(THF)}] + G_{\text{corr}}[\text{def2-SVP}]$, quasi-rigid-rotor harmonic oscillator at 298.15 K and 1 atm. A convergence adversary requires exactly one imaginary mode along the forming carbon-carbon bond rather than a spurious low-frequency wag, and the campaign, run on SDSC Expanse, hardened the harness against real cluster failures (an ORCA/SLURM library path, open-shell SCF stabilization on the magnetic nickel center).

Results. The experimental study reports $> 99\%$ enantiomeric excess, assigns the (*S*) product crystallographically, and, in the paper and across its 356-page supporting information, computes no transition state, offering instead a qualitative model that blames the minor pathway on a steric clash between the substrate methyl and the ligand backbone [76]. Three things the pipeline produced without further instruction are what this case reports: the free-energy surface, a coordination isomer nobody supplied, and a correction to the experimentalists’ cartoon.

The surface. From the single supplied intermediate the system located the reductive-elimination saddle on all four *R/S* $\times$ *cis/trans* channels, each cleared by the convergence adversary (Fig. 4c). On the oxygen-coordinated manifold *trans-S* is the lowest, 17.2 kcal mol⁻¹ above separated reactants ($\Delta G^\ddagger = 17.3$ kcal mol⁻¹) and 7.4 kcal mol⁻¹ below the lowest *R* saddle. *S* is predicted, the observed enantiomer, by a margin consistent with $> 99\%$ ee.

The isomer. The conformer search surfaced a nitrogen-coordinated binding mode it had never been given, and that mode is the *more stable* intermediate, below its oxygen-coordinated counterpart by 1.4 and 5.0 kcal mol⁻¹ on the *R* and *S trans* channels. A workflow that stopped at thermodynamics would have routed the reaction through it. Its reductive-elimination barriers are nevertheless higher (23.4 and 25.3 vs. 17.3 kcal mol⁻¹) and its saddles lie above oxygen-coordinated *trans-S* (18.3 and 18.7 vs. 17.2 on a common reference), so the deeper well is a thermodynamic sink, not the reactive species (Fig. 4d). What makes the check worth reporting is that the search, not the researcher, raised the alternative, and that it is settled by the barrier rather than by the well depth, a distinction an autonomous system has to be built to make.

The origin. The system then climbed the selectivity-analysis ladder on the matched *trans* pair, which differs only at the stereocentre (*R* at 29.9 vs. *S* at 17.2; $\Delta\Delta G_{\text{matched}}^\ddagger = 12.7$ kcal mol⁻¹), and each rung overturned the reading of the one below it. A noncovalent-interaction map and a buried-volume analysis both flag the expected clash, in the minor saddle the nearest methyl hydrogen sits 2.46 Å from a ligand aryl carbon, 0.44 Å inside van der Waals contact, against 3.23 Å in the major one, but neither is an energy. An activation-strain analysis [79] then assigns only -0.9 kcal mol⁻¹ to fragment distortion, a small term, and of the wrong sign, so the gap is carried entirely by the interaction between the chiral ligand and the nickel core; and a sobEDA decomposition [80] of that interaction finds the Pauli repulsion *lower* in the minor saddle (-1.6 kcal mol⁻¹), with the electrostatic and orbital terms, the two components of the Ni←N dative bonds, carrying 96% of the discrimination (Fig. 4e). The geometry agrees: in the minor saddle the ligand’s Ni–N bonds lengthen while *every* core bond to nickel shortens, so the ligand is levered off the metal rather than released by a more reduced one. The clash the experimentalists drew is real, and it is the *cause*; it is not the *cost*. Reaching that distinction took four analyses across three programs, none of them requested, which is the point of the harness: a geometric proxy can locate a contact, and only the ladder above it can price one.

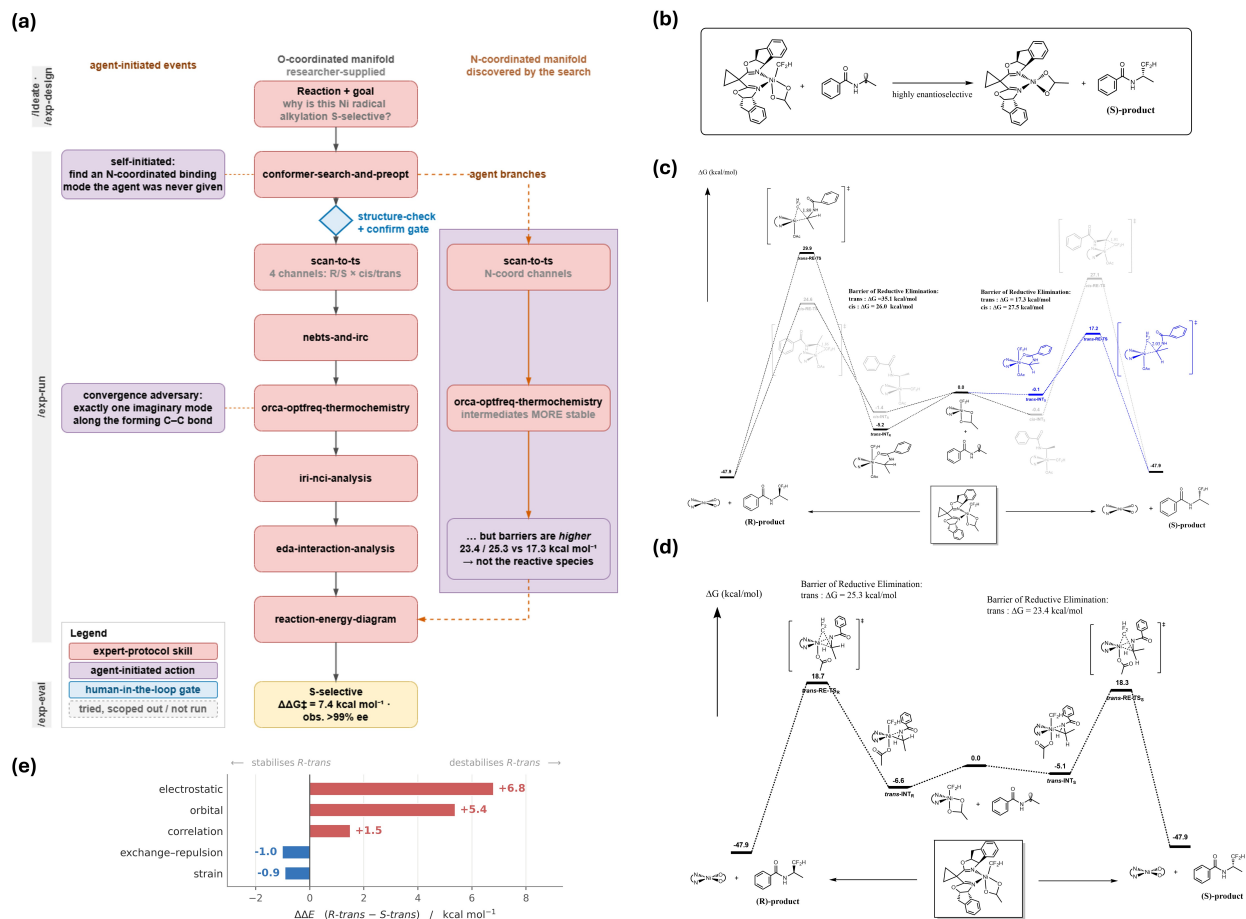


Figure 4: Enantioselectivity of a nickel-catalyzed asymmetric radical alkylation. **(a)** The campaign as executed: the centre column is the pipeline, one box per expert-protocol skill; the far-left gutter marks the lifecycle stage; purple boxes are actions the agent initiated on its own; the blue diamond is the human-in-the-loop structure-check gate; and the orange branch is the coordination-isomer check. **(b)** The reaction. Nickel-carbon coupling (step 1) precedes carbon-carbon reductive elimination (step 2); for this class of couplings the capture is fast and reversible while the reductive elimination is enantiodetermining [77, 78], so only step 2 is modelled. **(c)** Oxygen-coordinated free-energy surface over the four *R/S* × *cis/trans* channels. *trans-S* is the lowest saddle (17.2 kcal mol⁻¹ above separated reactants; $\Delta G^\ddagger = 17.3$ from its own intermediate), 7.4 kcal mol⁻¹ below the lowest *R* saddle, so *S* is predicted, the observed enantiomer. **(d)** The nitrogen-coordinated surface on the *trans* channels. Its intermediates are the *more stable* species (by 1.4 and 5.0 kcal mol⁻¹), yet its reductive-elimination barriers are higher (23.4 and 25.3 vs. 17.3) and its saddles lie above *trans-S* (18.3 and 18.7 vs. 17.2): a thermodynamic sink, not the reactive species. Panels (c) and (d) share one reference (separated reactants = 0.0, products = -47.9 kcal mol⁻¹). **(e)** Decomposition of $\Delta\Delta G^\ddagger_{\text{matched}} = 12.7$ kcal mol⁻¹ between the matched *trans* saddles, which differ only at the stereocentre; positive destabilises *trans-R*. Exchange-repulsion is *lower* there (-1.0; the Pauli component alone is -1.6), and the electrostatic and orbital terms, the two components of the Ni←N dative bonds, carry the discrimination. Strain comes from an activation-strain analysis, the other four terms from a sobEDA decomposition on the same fragmentation.

2.5 A validate-then-discover virtual screen for SARS-CoV-2 main-protease inhibitors

Motivation. Structure-based virtual screening is the highest-throughput stage of early drug discovery, but a screen is only trustworthy if its enrichment is demonstrated on known actives *before* it is turned loose on unknowns. This campaign, the system’s clearest excursion into applied biomolecular screening, runs the full lifecycle on the SARS-CoV-2 main protease (Mpro), a validated antiviral target: it first proves the pipeline enriches known binders, then prospectively screens a fresh drug-like library with no answer key, and triages the resulting hits honestly. It exercises a docking and cheminformatics engine stack (AutoDock Vina, RDKit, ProLIF) entirely distinct from the density-functional and molecular-dynamics stacks used elsewhere, and pairs with the binding-free-energy case below to show two complementary faces of the same domain.

Agentic workflow and lifecycle. From an idea and experiment-design page, the pipeline prepares the receptor (protonation and a docking box around the catalytic dyad), self-docks the native crystallographic ligand as a positive control, benchmarks retrospective enrichment against physicochemically matched decoys, and only then executes a 50,000-molecule prospective screen followed by a four-lens hit triage: active-band docking score, chemotype novelty, canonical pocket engagement, and a pan-assay-interference (PAINS) / structural-alert filter. The skills exercised, `protein-prep`, `molecular-docking`, `binding-affinity-eval`, `novelty`, `rdkit-featurization`, `nature-figure`, trace the whole idea-to-figure conveyor; an MM-GBSA endpoint ablation was tried and dropped as marginal, leaving a defensible docking-only pipeline. Every structure-generating and submission step passes the human-in-the-loop gates; the docking and scoring campaign ran on SDSC Expanse.

Results. The methodological controls pass first (Fig. 5). Self-docking reproduces the crystal pose of the native ligand to 0.264 Å heavy-atom root-mean-square deviation, so the docking function can recover a known answer (Fig. 5d); the decoy set is physicochemically matched to the actives in molecular weight and cLogP, so enrichment cannot be an artifact of the two populations simply looking different (Fig. 5e); and the inexpensive docking-exhaustiveness setting used for the screen reproduces the thorough one within noise (Spearman $\rho = 0.991$). Only then is enrichment measured: on the retrospective benchmark the pipeline ranks known Mpro actives with a receiver-operating-characteristic area under the curve of 0.795 (Fig. 5f) and enrichment factors of 16.5 $\times$, 7.7 $\times$, and 5.0 $\times$ in the top 1%, 5%, and 10% of the library (Fig. 5g). Turned prospective on 50,000 novel ZINC molecules with zero overlap with the validation library, 49,404 docked successfully and the four-lens funnel yields 183 three-lens candidates and 100 interference-clean hits (best -10.1 kcal mol $^{-1}$; Fig. 5a, h), whose contact frequencies reproduce the canonical Mpro pocket (Gln189, Met165, Glu166, His163; Fig. 5c). The campaign reports its own caveats rather than a headline gallery: pocket engagement is predominantly hydrophobic, with directional hydrogen bonds in the minority (His163 24%, Glu166 17%), and the shortlist skews to high molecular weight, a known docking size bias, both of which the confirmatory tier should weight.

2.6 Reconstructing the proton-supply mechanism of an isolated-Pd CO-reduction catalyst at the electrified interface

Motivation. An experimental study reports that atomically isolated Pd in copper promotes the electrochemical reduction of CO to acetate, and assigns the selectivity to isolated Pd acting as a *proton-supplying* site, a centre that activates interfacial water and thereby lowers the rate-determining protonation $\text{*CO} + \text{H}_2\text{O} + e^- \rightarrow \text{*COH} + \text{OH}^-$ [81]. That mechanism is read from electrochemistry

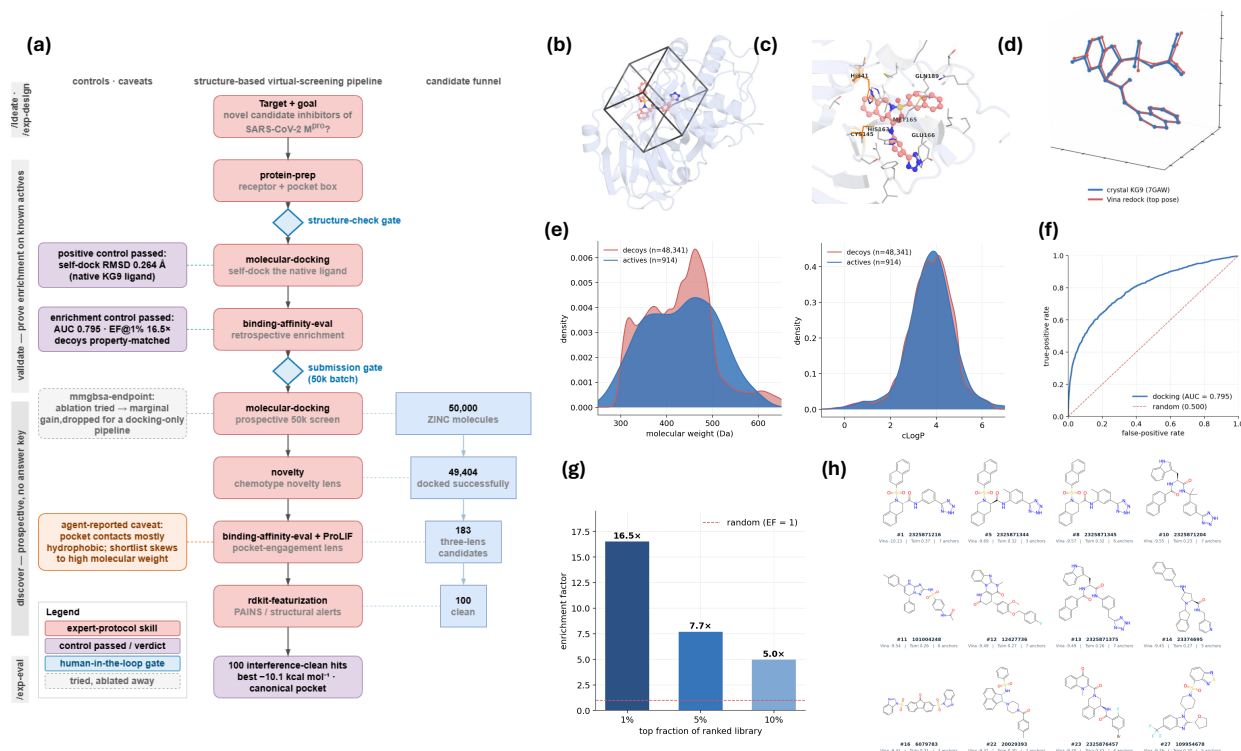


Figure 5: Validate-then-discover virtual screen on the SARS-CoV-2 main protease (M^{Pro}). **(a)** The campaign as executed: centre, the screening pipeline, one box per expert-protocol skill; left, the controls that passed and the caveats the system reported on itself; right, the candidate funnel; far left, the lifecycle stage. Blue diamonds are the human-in-the-loop gates. The campaign is organized so that enrichment is *proved on known actives* before any unknown is screened. **(b)** The prepared receptor with the docking box on the catalytic site. **(c)** The rank-1 prospective pose, engaging the canonical M^{Pro} residues (His41, Cys145, His163, Met165, Glu166, Gln189). **(d)** Docking is validated before it is trusted: re-docking the native crystallographic ligand (KG9, PDB 7GAW) reproduces the experimental pose to 0.264 Å heavy-atom RMSD (blue, crystal; red, top Vina pose). **(e)** The decoys are property-matched, molecular weight (left) and cLogP (right) of the 914 actives and 48,341 decoys overlap closely, so enrichment cannot be a physicochemical artefact. **(f)** Retrospective enrichment on that matched set: area under the receiver-operating characteristic 0.795, against 0.500 for random ranking. **(g)** Enrichment factors of 16.5x, 7.7x, and 5.0x in the top 1%, 5%, and 10% of the ranked library. **(h)** Only then are unknowns screened: the twelve interference-clean candidates, annotated with Vina score, Tanimoto similarity to the nearest known active, and the number of canonical pocket anchors engaged.

and spectroscopy, not resolved directly, and it lives at a place simulation rarely reaches cleanly: the electrified metal–water interface, where a *structural* claim (Pd reorganises the interfacial hydrogen-bond network), an *energetic* one (Pd lowers the water-dissociation barrier) and an *electronic* one (why) are one and the same story. This campaign reconstructs that story from the paper alone, building the explicit-water constant-potential interface and testing the proton-supply hypothesis across all three scales (research mode A).

Agentic workflow and lifecycle. From the source paper the agent reduced the claim to a falsifiable chain, a more connected, dynamic interfacial water network $\rightarrow$ easier water dissociation $\rightarrow$ easier CO protonation, and ran a seven-skill pipeline (Fig. 6a): literature grounding; slab construction with single-atom substitution, an explicit-water film and adsorbate placement, giving Cu(111) $\pm$ isolated Pd $\pm$ *CO with 44 explicit waters; *ab-initio* molecular dynamics for the interfacial structure and dynamics; constant-potential slow-growth thermodynamic integration of the water O–H coordinate, with the Fermi level calibrated to the experimental potential (-0.7 V vs. RHE, 2 M KOH $\Rightarrow \mu = -2.90$ eV); static single-points for the projected d-band centre and Bader charges; and figure assembly. Throughout, the potential, the explicit-water film and the driven collective variable were configured to match the reported electrochemistry rather than a gas-phase idealisation, and the charge analysis was grounded to the adsorbate-oxygen trend consistent with the parent work’s spectroscopy.

Results. The campaign reconstructs the proton-supply mechanism as a single, coherent multi-scale picture (Fig. 6b). The interfacial water network is the through-line: on Cu–Pd the first water layer is markedly more connected (hydrogen-bond density peak ≈ 4.2 vs. ≈ 3.0), it stays denser around the adsorbed *CO (3.95 vs. 3.72 hydrogen bonds per interfacial water), and it turns over faster (intermittent hydrogen-bond correlation time $\tau = 1.55$ vs. 1.65 ps), a network that is at once denser and more labile, the combination an efficient Grotthuss-type proton relay needs. That network dissociates water more readily: the constant-potential slow growth places the barrier lower at the Pd site ($E_a \approx 1.41$ vs. 1.73 eV at -0.7 V), with the liberated proton released into solution rather than onto the metal, the proton-supply signature made explicit. The electronic origin closes the loop: the Pd d-band centre sits nearer the Fermi level than Cu’s (-2.23 vs. -2.30 eV), and the dissociated *OH oxygen carries less excess charge at Pd (-0.99 vs. -1.11 e), a stronger, more covalent Pd–OH interaction that stabilises the proton-supply product, the same charge-modulation lever the parent work invoked for the *COH–*CO intermediate, here extended to the upstream water-activation step. The reconstructed chain recovers the experimental hypothesis end-to-end across structure, dynamics, energetics and electronic origin; a constant-potential *CO $\rightarrow$ *COH barrier is the natural next step to close it on the product side.

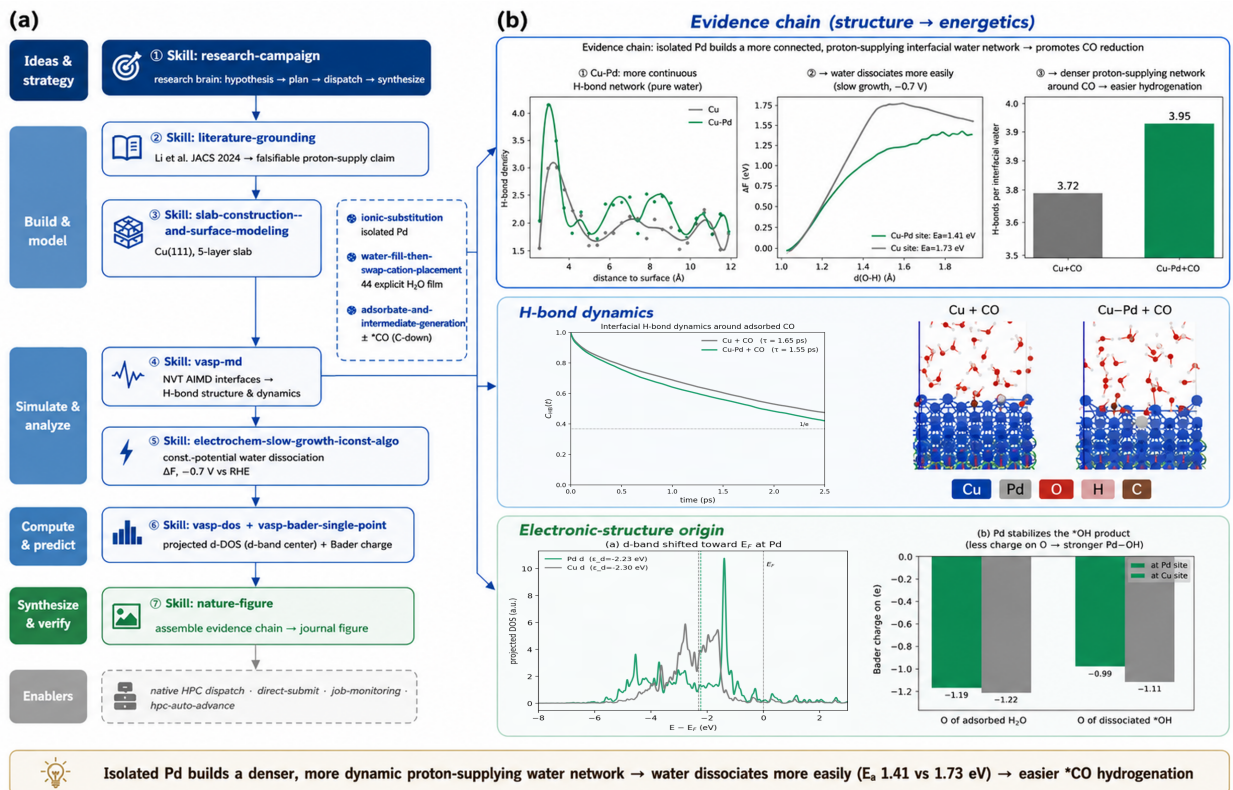


Figure 6: An isolated-Pd CO-reduction mechanism reconstructed end-to-end at the electrified interface. **a**, The seven-skill pipeline, grouped by research stage (ideation → build → simulate → compute → synthesize) over a native HPC-dispatch substrate. **b**, The evidence chain. *Top*: the first interfacial water layer is more connected on Cu-Pd (hydrogen-bond density peak ≈ 4.2 vs. ≈ 3.0); constant-potential slow growth places the water-dissociation barrier lower at the Pd site ($E_a \approx 1.41$ vs. 1.73 eV, -0.7 V); the network stays denser around *CO (3.95 vs. 3.72 per interfacial water). *Middle*: the intermittent hydrogen-bond autocorrelation around *CO ($\tau = 1.55$ vs. 1.65 ps) with interface snapshots. *Bottom*: the projected d-band centre (-2.23 vs. -2.30 eV) and the Bader charge on the oxygen of adsorbed H_2O and dissociated *OH (less excess charge at Pd, -0.99 vs. -1.11 e). Hydrogen-bond criterion: $O \cdots O < 3.5$ Å, $\angle O-H \cdots O > 120^\circ$.

3 Discussion

ChemMate occupies an intersection that the surveyed systems leave empty, but its design trades autonomy for reliability in ways that bound what it claims, and we state those bounds plainly.

The underlying language models are non-deterministic, so the same prompt does not always yield the same plan. We treat this as a measurement problem rather than a defect to hide: quantifying the run-to-run spread by repeating protocol tasks is a natural control we leave to future work, and the runtime contract already narrows the blast radius of a bad sample by refusing graph and field writes that violate the schema. Non-determinism is the reason convergence is judged by force rather than trusted, and the reason the human gates exist where they do.

Several engines ChemMate drives are licensed, VASP among them, and a campaign that depends on a licensed code cannot run on a cluster that lacks the license. The skill library is deliberately engine-plural, open-source plane-wave and Gaussian-basis codes, machine-learning potentials, and semi-empirical methods are all reachable, so that intent can usually be served by an available engine, but the dependence is real and we do not claim license-free portability. When a preferred engine is unavailable and an alternative is substituted, the system says so explicitly, because basis sets,

pseudopotentials, and convergence criteria do not transfer between codes.

The gates that make supervision operational also cost latency. A human-in-the-loop checkpoint at every structure and every submission is, by construction, slower than full autonomy, and a reviewer who measures a system by its degree of unattended operation will see ChemMate as deliberately less autonomous than systems that submit without asking. This is the intended trade. The costliest mistake in high-performance computing is a wrong input run at scale, and a gate that catches it pays for its latency in allocation not burned. The mobile loop exists to shrink that latency to the time it takes to glance at a phone, not to remove the human.

The system is coupled to a single coding-agent harness, and that coupling is a limitation: portability to a different harness is not demonstrated here, and some behavior depends on the host harness’s tool model. The skills and the runtime contract are the durable, portable part; the harness binding is the part most likely to change.

An obvious objection deserves a direct answer: could the bare coding agent, without ChemMate, not simply do this? Partly, and sometimes, frontier models carry real chemistry knowledge, and we expect the unaided harness to clear some protocol tasks. The natural control is a scaffold ablation: the same tasks, the same model, one arm stripped of every skill, contract, and gate. Such an ablation would measure three quantities the anecdotes cannot: the reliability gap across repeated runs, the rate at which the unaided model falls into named domain traps, pseudopotential ordering, slab-versus-bulk k-point choices, a scheduler’s exit code mistaken for convergence, and the auditability gap, which no amount of unaided competence closes. An unaided success leaves no receipt, no gate trail, and no replayable record; in a domain where a result is defensible only from its audit trail, an unauditable success is not yet a result. We pre-register this ablation and report it as the natural next step; strong unaided performance on protocol tasks would sharpen, not weaken, the claim that the durable contribution is orchestration and auditability rather than tacit chemistry knowledge.

These limitations also answer, by construction, the skeptic’s charge that current agents are co-scientists rather than autonomous discoverers [56]. ChemMate does not claim to remove the human from problem selection or success-criterion setting; it keeps the human in the loop on purpose. The skeptic’s specific calls are met in the architecture: the demand for simulators as verifiers and physical feedback is answered by a substrate that is physical simulation with force-based convergence checks and adversarial verification, and the demand for persistent world models is answered by the typed knowledge graph and the receipts ledger, which are designed to make a campaign a durable, replayable record rather than a transcript. We position ChemMate not as an autonomous discoverer but as an instrument that automates the lifecycle while leaving judgment where it belongs.

The agentic automation of science has so far split along a line: systems that reason their way from an idea to a manuscript do not run physical simulation, and systems that run physical simulation do not carry a project past the calculation. ChemMate joins the two for computational chemistry. It carries a project from an idea through a gated, multi-cluster high-performance-computing campaign to a manuscript draft, with an orchestration layer built to leave a receipt per job and a typed graph from which the whole can be replayed, and it keeps a human in the loop where wrong inputs would waste allocation, actionable from a phone. The contribution is not a faster calculation. It is the demonstration that the unit of automation can be the research lifecycle rather than the calculation, in a domain where the simulation is real, the hardware is heterogeneous and spread across institutions, and a result is worth nothing until it can be defended from its own audit trail. The campaigns reported here, spanning four scientific domains and ranging from a machine-learned Hamiltonian that corrected its own success metric to a mechanism reconstructed, from a published paper alone, across three scales, each defended from its receipts and knowledge graph, show what that looks like in practice.

4 Methods

4.1 Runtime contract and skill governance

The wiki engine and its runtime contract derive from the open-source AutoSci platform (MIT) [6] and are extended rather than rebuilt: the typed-page schema gains the compute-project fields (receipts, research modes), and the idea-pipeline skills gain the chemistry branches described below. A per-file provenance inventory (12% inherited unmodified, 7% modified, 81% native by file count) ships with the code. The skill library is kept coherent by a declarative, machine-checked runtime contract. Schema files fix the shape of every page and edge in the knowledge graph, a policy file records which skill may write which field, and a cross-reference specification requires that every forward link be matched by its reverse. A validator gates the metadata write path, an unknown field or an off-enum value is rejected at the write, not discovered later, and a repository lint sweeps every page, edge, and cross-reference after each ingestion or graph mutation, so a skill cannot silently corrupt the graph or invent a field. The contract also governs how skills evolve: a durable lesson from a campaign is folded back into the owning skill only through a human-reviewed proposal step, never by an automatic self-edit. This keeps self-improvement under explicit control, a deliberate contrast to unsupervised optimization.

4.2 Gap-driven ideation

Ideation reads the wiki before it reads the web. A landscape scan loads the graph’s compressed context, its open-question map, the open problems recorded on concept and topic pages, the list of active ideas, and a banlist of failed ideas with their failure reasons; external literature search is weighted by wiki maturity, heavier on a cold wiki, lighter once the graph is dense. The researcher may seed a round explicitly with a direction or with seed papers, which are ingested on the fly; absent a direction, the scan selects the most valuable open question. Candidate ideas are then generated along five structured paths bound to typed schema fields, landscape-driven, incremental on a method’s recorded limitations, combination of two methods’ trade-off profiles, relaxation of an assumption shared across methods, and cross-domain transfer by mechanism similarity, by two language models brainstorming independently, whose candidates are merged and deduplicated. Each idea declares its research mode and the reference it will be judged against (mode A: an experimental or higher-theory value; mode B: a benchmark set plus a baseline method; mode C: the limiting case the derivation must recover); an idea that cannot name its reference is pushed back. Survivors pass a mode-aware feasibility check, core-hours on the target cluster, the cost of a benchmark sweep, or analyst time, then a novelty search and a scored review; the researcher confirms the final selection, and every idea, including eliminated ones with their reasons, is written to the graph so that failure feeds the next round’s banlist.

4.3 Project layout, receipts, and multi-cluster submission

Directory names are human-readable and describe what a calculation is, a relaxation of a hydrogen-covered copper surface is `Cu111-H_relax`, not a hash or a timestamp, and a rerun takes a readable version suffix rather than a new opaque name. Volatile metadata (date, job identifier, status, provenance) lives in a per-calculation `_meta.yaml` and in pointer indices, not in the name, so a directory stays named for its science. Every calculation directory lives inside a `calc/` container so that a screen of hundreds of candidates cannot flood the project root, which holds only typed directories and the project’s own narrative files.

Each job leaves a durable receipt: its actual remote location, the machine, host, and remote directory, is persisted in `_meta.yaml` and aggregated into a project-root `_REMOTES.md` ledger, so a project that spans several clusters at different institutions stays traceable, and the answer to “which calculation ran where” survives a change of preset or remote root. Remote directories mirror the local layout, so the audit trail is identical on both sides. Throughput does not explode the index: a single calculation submits through one path, while a multi-candidate screen submits through a batch path that produces one batch directory, a per-candidate manifest, and a single rolled-up index and ledger row, rather than one row per job. The layer monitors running jobs and advances each finished candidate independently so that a slow member never blocks the rest. Every collected outcome passes a failure classifier, out-of-memory, walltime, an unconverged SCF, a node fault, a transient infrastructure error, whose category lands in the calculation’s receipt, and a decay-aware retry policy bounds how many times, and with what mandatory change, a classified failure may be resubmitted. Each project owns its cluster configuration in a project-local preset that submission reads in preference to any user default, and a verification step probes the connection, scheduler, remote root, and dependency paths over a remote session and stamps the result, so a wrong path is caught before it fails every job rather than after.

The same layout supports mid-stream takeover. An existing project that predates ChemMate is brought under the layout by an adoption pass that imports each calculation into the `calc/` container under a readable name, reconstructs its receipt from whatever provenance survives, and marks it as imported in the project index; the remote side is reorganized to mirror the local tree only through a plan-then-approve tidy step that records every move in a manifest, never by hand. Each project additionally carries narrative resume files, a TL;DR-first README, a plan, and a lessons ledger, kept current as the project advances, so a fresh session, human or agent, can take the project over from disk at whatever stage it is in.

4.4 Gates, cancellation safety, and convergence verification

Every structure-related operation, slab construction, adsorbate placement, batch structure generation, supercell construction, substitution, defect creation, and every submission whose inputs include a structure first asks the user whether to check or modify the structure. For a surface cut, both the source bulk and the resulting slab are staged in the calculation directory, so that a user who judges the cut wrong can re-cut from the bulk rather than being stuck with the agent’s slab. The policy is enforced, not merely recommended: a pre-execution hook hard-blocks the mutating call, structure generation, pseudopotential assembly, or submission, until an explicit confirmation flag is set, so the gate blocks and the structure check makes the confirmation informed. Cancellation carries its own safety rule: a job is cancelled only by its explicit identifier, never by user, account, or partition, because an agent session may share an allocation with unrelated work.

A calculation’s result is not trusted on the scheduler’s word. An adversarial convergence verifier inspects a finished calculation and returns one of a small set of verdicts, converged, not converged, unphysical, or not checked, judging convergence by force rather than by the scheduler’s exit code. The verifier is invoked before a result is written or reported and is given only the bare claimed verdict, not the claimant’s reasoning, so it cannot be anchored by the very argument it is meant to check.

4.5 The mobile supervision loop

Structures and input files surface in CatGo, our previously published cross-platform structure, workflow, and campaign-orchestration workbench [58, 59], used here as infrastructure rather than as

a contribution of this paper. CatGo runs on phone, tablet, and desktop (Windows, macOS, Linux) alike; its client unifies, on one surface, a remote connection with an embedded terminal, structure visualization, and structure and input-file editing, so a researcher can open an agent-proposed slab on a phone, re-cut a surface or correct an adsorbate site or adjust an input deck on the live workstation, and then approve the gated submission, without a desk. The corrected artifact re-enters the gated pipeline. The novel object is not remote access on a phone, which is commonplace, but the unified closed loop: agent orchestration, cluster receipts, structure visualization, live edit, and gate approval integrated on one surface, available down to a phone (Fig. 7). To our knowledge, among autonomous simulation-agent systems, none closes the research loop off the workstation in this way.



Figure 7: The mobile supervision loop. The human-in-the-loop cycle runs on a single CatGo client, down to a phone: the agent proposes a structure and input deck (1), they surface on one client (2), the human inspects and edits on the live workstation (3, re-cut a surface, correct an adsorbate site, adjust an input deck), approves the gated submission (4), and the corrected artifact re-enters the gated pipeline. The novel object is the unified closed loop on one surface, not remote access itself.

4.6 Knowledge-graph schema

Pages are typed, ideas, experiments, results, methods, concepts, people, papers, and edges between them carry direction, attributes, and confidence, with citations attached as first-class records. Because every forward link is matched by its reverse and validated against the contract, the graph is a navigable record of how a project moved from a question to a written claim. The same graph supplies the related-work corpus and citation plan, so the literature a project rests on is part of its replayable record. Shipped alongside the project tree, the graph and the receipts together form an audit trail from which any reported result can be reconstructed.

Data availability

The data supporting the findings of this study, the per-campaign inputs, converged outputs, parsed result tables, and the knowledge-graph records, are organized under the project layout described in Methods and are available from the corresponding author on reasonable request. A public deposit of the released campaign data and the knowledge-graph export will accompany publication.

Code availability

ChemMate is built on a general-purpose coding-agent harness together with a machine-checked runtime contract and a library of expert-protocol skills, as described in Methods. The runtime contract, skill definitions, and the project- and HPC-layout tooling are available from the corresponding author on reasonable request, and an archived release will accompany publication.

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Author contributions

All authors contributed to the design of the ChemMate system, the specification and execution of the computational campaigns, the analysis of the results, and the writing and revision of the manuscript.

Competing interests

The authors declare no competing interests.

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